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MINI PROJECT REPORT

on

“CHEST X-RAY ANALYSIS TO DETECT PNEUMONIA”

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B. M. S. COLLEGE OF ENGINEERING

(Autonomous Institution under VTU)

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(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the project work entitled “CHEST X-RAY ANALYSIS TO DETECT PNEUMONIA” carried out by **AAKANKSHA V R (1BM22CS001), ABHINAV RAGHU(1BM22CS005), AISHWARYA VIJAY(1BM22CS025) AND ARYA HIMANSHU (1BM22CS055)** who are bonafide students of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraiiah Technological University, Belgaum during the year 2023-2024. The project report has been approved as it satisfies the academic requirements in respect of **Mini Project (23CS5PWMIP)** work prescribed for the said degree.

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DECLARATION

We, AAKANKSHA V R(1BM22CS001), ABHINAV RAGHU(1BM22CS005), AISHWARYA VIJAY (1BM22CS025), ARYA HIMANSHU(1BM22CS055), students of 5th Semester, B.E, Department of Computer Science and Engineering, B. M. S. College of Engineering, Bangalore, here by declare that, this Project Work-1entitled " CHEST X-RAY ANALYSIS TO DETECT PNEUMONIA " has been carried out by us under the guidance of Prof. Namratha M , Assistant Professor , Department of CSE , B. M. S. College of Engineering, Bangalore during the academic semester September 2024- January 2025.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

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1.Introduction

Pneumonia is a serious health issue that affects millions of people worldwide, particularly children and the elderly. Early detection is key to ensuring timely treatment and preventing complications, but the traditional ways of diagnosing pneumonia—like physical exams, chest X-rays, and manual reviews by radiologists—can be slow, costly, and sometimes inaccurate. These limitations often lead to delays in treatment or missed diagnoses.

Thankfully, modern technology offers a way forward. Tools like artificial intelligence (AI) and machine learning (ML), especially Convolutional Neural Networks (CNNs), are changing how we approach medical imaging. CNNs are excellent at analyzing images and can be used to detect pneumonia from chest X-rays quickly and accurately.

This report dives into how CNNs can make pneumonia diagnosis faster and more reliable. By combining CNN technology with imaging tools and cloud systems, we can create solutions that not only improve patient care but also make healthcare more efficient and accessible for everyone.

1.1 Motivation

The motivation behind this project arises from the growing challenges in diagnosing pneumonia effectively, especially in resource-limited settings where delays or inaccuracies in detection can have severe consequences. Despite the critical need for timely and accurate diagnosis, many healthcare systems still rely on traditional methods that are not only time-consuming but also prone to human error. This often results in delayed treatment and worsened patient outcomes.

With the increasing burden of pneumonia globally, particularly in vulnerable populations like children and the elderly, there is an urgent need for a technology-driven solution. By leveraging automated detection systems using Convolutional Neural Networks (CNNs), this project aims to empower healthcare providers with tools to identify pneumonia early, improve diagnostic accuracy, and enable prompt treatment, ultimately enhancing patient care and outcomes.

1.2 Scope of the Project

- Evaluating traditional and modern methods for pneumonia diagnosis.
- Implementing Convolutional Neural Network (CNN) models to analyze and detect pneumonia based on analysis of x-ray images.
- Developing a system that utilizes image processing techniques for accurate and efficient diagnosis of pneumonia.
- Investigating the potential integration of additional clinical data (e.g., patient symptoms, environmental factors) to enhance diagnostic accuracy.

- Recommending effective treatment pathways based on the severity and classification of detected pneumonia.

1.3 Problem statement

The scope of this project focuses on tackling the challenges of pneumonia detection using modern technology:

- Understanding the different types of pneumonia and how they appear on chest X-rays.
- Comparing traditional diagnostic methods with AI-powered approaches to highlight their strengths and limitations.
- Building and training Convolutional Neural Network (CNN) models to accurately analyze chest X-rays and identify pneumonia.
- Developing an automated system that simplifies and speeds up the detection process, reducing the burden on radiologists.
- Exploring cloud-based tools to make these diagnostic systems accessible to healthcare providers in remote or resource-limited areas.
- Offering practical insights and recommendations to improve pneumonia diagnosis and treatment based on the results from this system.

Aim of the project is to bridge the gap between traditional medical practices and cutting-edge technology, making pneumonia detection faster, more accurate, and widely available.

2.Literature Survey

2.1)Detection of Pneumonia Using Convolutional Neural Networks

Methodology:

- Utilized Convolutional Neural Networks (CNNs) for automated feature extraction and classification.
- Implemented a boosting strategy to enhance classification performance while reducing computational complexity through dimensionality reduction.
- Employed Scale-Invariant Feature Transform (SIFT) for detecting key feature points in chest X-ray images.
- Focused on Region of Interest (ROI) segmentation, specifically targeting lung regions, to improve classification accuracy.
- Applied K-means clustering and Bag-of-Visual Words (BoVW) for feature clustering, reducing the computational burden.

Pros:

- Reduction in Computation: Dimensionality reduction techniques significantly decreased computation complexity.
- Improved Accuracy: Achieved higher classification accuracy compared to existing CNN-based models.

Cons:

- Limited Generalization: Models like MobileNetV2 and Xception struggled with convergence, showing poor classification performance.
- Computation Power: Some CNN models required substantial computational resources, limiting their applicability in resource-limited environments.

Results:

- Achieved a validation accuracy of 80.1% and an F1 score of 80.1%, outperforming traditional machine learning approaches.
- Noted that models such as MobileNetV2 and Xception failed to converge during training process.

2.2)Detection of Pneumonia Using Convolutional Neural Networks and Transfer Learning

Methodology:

- Utilized Convolutional Neural Networks (CNNs) combined with transfer learning techniques for pneumonia detection.
- Employed models such as ResNet, AlexNet, and VGG for deep feature extraction from chest x-ray images.
- Fine-tuned pre-trained models using transfer learning, leveraging weights learned from large image datasets.
- Focused on Region of Interest (ROI) segmentation to target specific lung areas, enhancing classification accuracy.
- Applied data augmentation to increase training data diversity and improve model generalization.

Dataset Used:

- The ChestX-ray14 dataset, containing labeled chest X-rays, including pneumonia-positive and negative cases.

Pros:

- High Accuracy: Achieved robust classifications with high sensitivity and specificity.
- Automation: Automated feature extraction eliminates the need for manual feature engineering, improving efficiency.
- Deep Learning Flexibility: Transfer learning enhances performance, especially with smaller datasets.

Cons:

- Overfitting Risk: Smaller or imbalanced datasets increase the likelihood of overfitting.
- Computational Cost: Training CNN models is resource-intensive and demands significant computational power.

Results:

- Achieved up to 93% accuracy in classifying pneumonia, with high sensitivity and specificity.
- The pre-trained CNN models like ResNet and VGG, coupled with transfer learning, significantly improved classification performance compared to traditional methods.

2.3) Detection of Pneumonia Using CheXNet Model: A Deep Learning Approach

Methodology:

- Utilized CheXNet, a 121-layer Convolutional Neural Network (CNN) model for analysis of medical images.
- Trained the model on the ChestX-ray14 dataset to identify pneumonia among 14 labeled thoracic diseases.
- Applied transfer learning techniques and end-to-end training to fine-tune the model for pneumonia detection in chest X-rays.

Pros:

- Radiologist-Level Performance: Outperformed radiologists in accurately identifying pneumonia in chest X-rays.
- Generalization Capability: Demonstrated strong generalization across various thoracic diseases.
- Deep Architecture Benefits: The 121-layer architecture enables precise feature extraction for enhanced accuracy.

Cons:

- High Computational Demand: Requires significant computational resources for training and inference.
- Practical Limitations: Performance metrics like F1 score may not fully translate to clinical settings.
- Overfitting Risk: The deep architecture increases susceptibility to overfitting without adequate regularization.

Results:

- F1 score of 0.435 for pneumonia disease detection, exceeding the average performance of radiologists.
- Demonstrated good precision making it a good solution for automated diagnosis in resource-limited environments.

2.4) Detection of Pneumonia Using Vision Transformers (ViTs)

Methodology:

- Applied Vision Transformers (ViTs) for pneumonia detection in chest X-rays, utilizing their capacity to process images as a sequence of patches.
- Replaced traditional convolutional layers with self-attention mechanisms to classify chest X-ray images.

Dataset Used:

- Leveraged publicly available datasets such as the NIH Chest X-ray and RSNA Pneumonia Dataset for training and evaluation.

Pros:

- **Global Feature Capture:** Effectively captures global dependencies in images, providing robust classification capabilities.
- **Scalability:** Performs well on large datasets with complex visual patterns, making it suitable for medical image analysis.

Cons:

- **Data Requirement:** Requires large datasets to achieve optimal performance due to its reliance on attention mechanisms.
- **Computational Demand:** ViTs are computationally intensive, requiring substantial resources for training.

Results:

- Vision Transformers have shown promising performance in pneumonia detection, rivaling traditional CNN-based methods.
- Their ability to process and classify complex medical images suggests their potential for broader applications in diagnostic imaging.

2.5)Detection of Pneumonia Using AlexNet and Feature Selection

Methodology:

- Utilized the AlexNet architecture comprising pooling, convolutional and fully connected layers in order to extract features from chest x-ray images.
- Applied data augmentation techniques, such as flipping and rotation, to address dataset imbalance.
- Employed the mRMR algorithm to reduce the number of features from 1000 to 100 for efficient processing.

Pros:

- **Computational Efficiency:** mRMR feature selection reduced computational cost significantly.
- **Dataset Handling:** Data augmentation effectively balanced the dataset, improving model generalization.

Cons:

- **High Resource Demand:** Training AlexNet requires substantial computational resources.
- **Overfitting Risk:** The model's depth may lead to overfitting without proper regularization techniques.

Results:

- Achieved an accuracy of 96.83% when AlexNet features were combined with a Linear Discriminant Analysis (LDA) classifier after feature selection.

2.6)Detection of Pneumonia Using VGG16 and Image Enhancement Techniques

Methodology:

- Image Enhancement: Applied three enhancement methods—Histogram Equalization (HE), Contrast Limited Adaptive Histogram Equalization (CLAHE), and Equalized Histogram Filter (EFF)— before feeding them into the model.
- VGG16 Model: VGG16 CNN architecture to diagnose pneumonia in chest X-ray images. The images were resized and the model being trained for 50 epochs.

Dataset Used:

- The dataset was sourced from Kaggle’s public chest X-ray collection, originally compiled by Kermany et al. It contains 3,151 images, categorized into two groups: normal and pneumonia. Expert doctors reviewed and confirmed the dataset’s accuracy. Training: 1,341 normal images and 1,186 pneumonia images
- Validation: 234 normal images and 390 pneumonia images

Pros:

- Improved Diagnosis in Low-Resource Areas: The enhancement methods make pneumonia detection more accurate, particularly in resource-limited settings.
- Handles Large Datasets: VGG16 is capable of processing large datasets efficiently.

Cons:

- Data and Computational Demands: Requires a substantial amount of data and computational power for training.
- Risk of Overfitting: The deep architecture of VGG16 may lead to overfitting without proper regularization.

Results:

- The model achieved 94.51% accuracy in detecting pneumonia from training images.
- When tested on new, unseen images from the validation set, the model achieved an 86.7% accuracy in detecting pneumonia.

2.7)Automated Pneumonia Detection in Chest X-Ray Images Using Deep Learning

Methodology:

- The program was designed with multiple layers , followed by additional layers to classify whether the image showed pneumonia or not.
- The model had been trained using a set of labeled images, adjusted using a second set of images, and finally tested on a new group to evaluate its performance.

Dataset Used:

- The dataset consisted of 5,863 X-ray images of children aged 1 to 5 from a single hospital. The images were high-quality and verified by medical experts. Each image was labeled as either "pneumonia" or "normal" and split into training and testing groups.

Pros:

- High Accuracy: The program successfully identified pneumonia with 97% accuracy.
- Improved Diagnostic Speed: It can assist doctors in diagnosing pneumonia faster and more accurately.
- AI-Driven Learning: The deep learning model learns to identify key features in the images, improving its detection capabilities.

Cons:

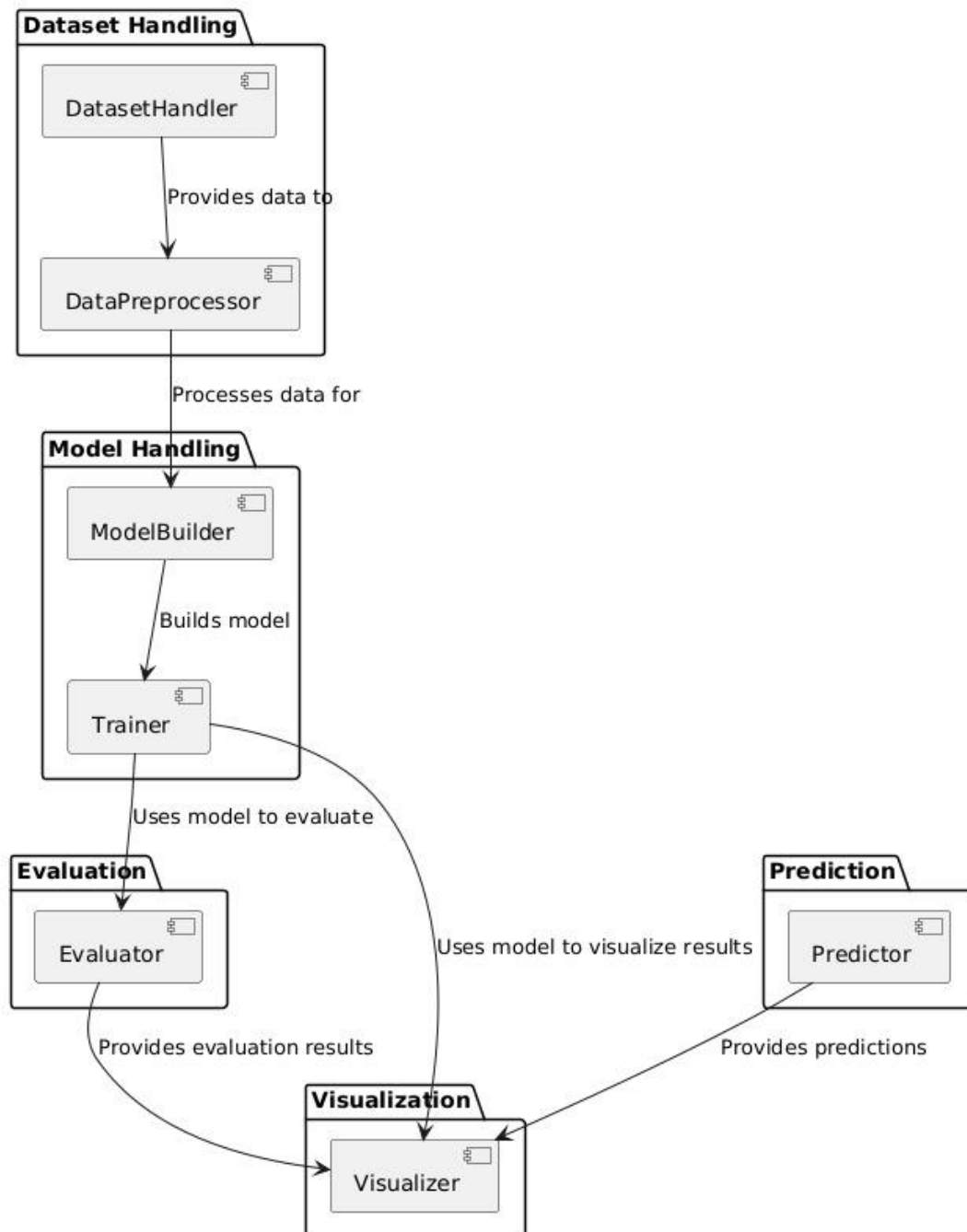
- Lack of Comparison: The study did not assess how the model compares to human doctors or other AI systems.
- Limited Generalization: The model might not perform as well with X-ray images from hospitals other than the one used in the study.

Results:

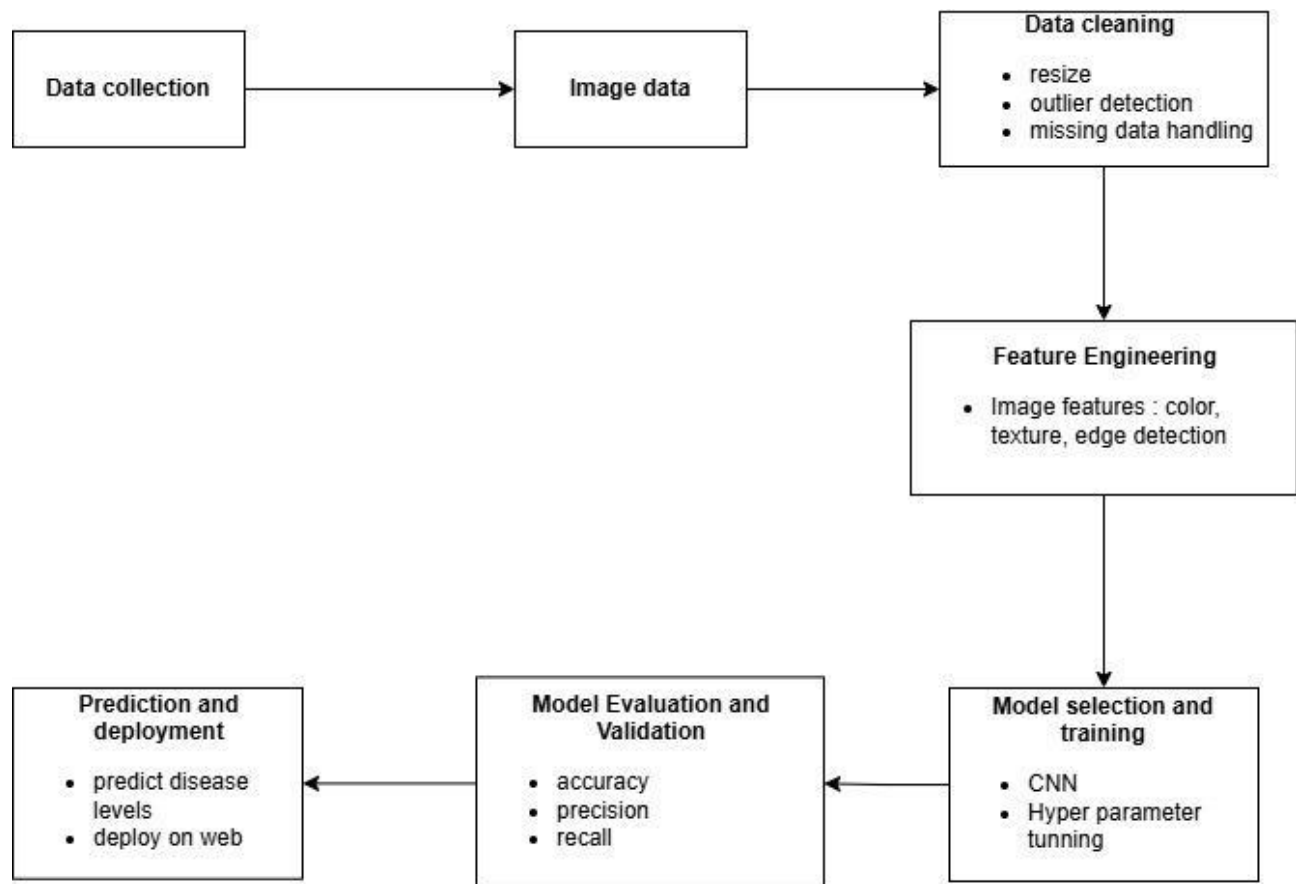
- The deep learning model demonstrated high effectiveness in identifying pneumonia in chest X-rays of young children, showing great potential as an aid for medical professionals.

3.Design

3.1 High Level Design



3.2 Detailed Design



4. Implementation

4.1 Proposed methodology

Data Collection:

We have used dataset comprising chest xray images classified as "Normal" and "Pneumonia."

Preprocessing:

Resize images to a standardized dimension of 224x224 pixels to comply with the input specifications of the MobileNetV2 architecture. Normalize pixel values to the interval [0, 1]. Employ data augmentation techniques (including rotation, zoom, and horizontal flips) to enhance model generalization.

Transfer Learning with MobileNetV2:

Employ MobileNetV2, which has been pre-trained on the ImageNet dataset, as the foundational model. Incorporate fully connected layers to adapt the model for binary classification task.

Training the Model:

Fine-tune the pre-trained model using the chest X-ray dataset. Implement binary cross-entropy as the loss function and the Adam optimizer for optimization process.

Testing and Evaluation:

Assess the model on a test set for performance metrics such as recall, F1-score, accuracy, precision.

Deployment:

We have designed an interactive application using Streamlit to allow users to upload chest xray images and get predictions.

4.2 Algorithm used for implementation

- **Preprocessing and Feature Extraction:** Commence by loading and resizing chest X-ray images to dimensions of 224x224 pixels, normalizing the pixel values, and employing the MobileNetV2 architecture, which has been pre-trained on the ImageNet dataset, to obtain salient characteristics from the images.
- **Dimensionality Reduction and Feature Learning:** Implement GlobalAveragePooling2D to diminish the dimensionality of the feature maps, succeeded by a Dense layer comprising 128 neurons with ReLU activation to effectively capture intricate patterns within the data.
- **Classification:** Incorporate an output layer consisting of a single neuron equipped with sigmoid activation to facilitate the classification of images as either "Normal" or "Pneumonia."
- **Optimization and Training:** Compile the model utilizing the Adam optimizer with binary cross-entropy loss and subsequently validate the model on the validation set to ensure precise classification of pneumonia cases.

4.3 Tools and technologies used

Programming Languages and Libraries:

The implementation will be conducted using Python, employing TensorFlow/Keras for the development of Convolutional Neural Network (CNN) models aimed at pneumonia identification through the analysis of chest X-ray images. OpenCV will be utilized for the preprocessing of images (including resizing and normalization) pertaining to chest X-ray data.

Frameworks and Tools:

NumPy and Pandas will serve as the primary tools for the handling, manipulation, and analytical assessment of the dataset. Flask will be employed to construct an interactive frontend that facilitates the visualization of outcomes and predictions related to pneumonia detection. MobileNetV2 will be incorporated as the transfer learning model to enhance feature extraction capabilities and overall performance.

Hardware:

Systems equipped with GPU capabilities (such as NVIDIA CUDA) will be utilized for training and of the models, which is particularly advantageous when dealing with extensive datasets like X-ray images

Deployment Tools:

Flask for creating an interactive user interface where users have to upload chest xray images and

obtain predictions for pneumonia detection in real-time.

4.4 Training

Dataset Preparation: Images are systematically classified into "Normal" and "Pneumonia" categories and subsequently divided into training, validation, and testing subsets. Data augmentation techniques are employed during the training phase to enhance variability and bolster the robustness of the model.

Training Configuration: Pre-trained Model: MobileNetV2 with its base layers initially retained in a frozen state. Custom Layers: Incorporated to facilitate binary classification. Hyperparameters: Optimizer: Adam with a specified learning rate of 0.0001.

Training Steps: Begin with the initialization of MobileNetV2 utilizing pre-trained weights sourced from ImageNet. Integrate and train custom layers utilizing the chest X-ray dataset. Fine-tune the foundational model by selectively unfreezing a portion of the layers to better accommodate the dataset. Implement class weights to rectify the imbalance present between the "Normal" and "Pneumonia" categories. Continuously monitor validation loss to mitigate the risk of overfitting.

4.5 Testing

Evaluation Metrics:

The trained model is evaluated using the test dataset to assess its performance on unseen data.

Accuracy:

A metric that quantifies the proportion of predicted positives that were indeed correct.

Precision:

The ratio of accurately classified images to the total number of images.

Recall (Sensitivity):

A metric that assesses the number of actual positives that were accurately identified.

Testing Process: Assess the performance of the trained model utilizing the test dataset. Construct a confusion matrix to illustrate the counts of true positives, true negatives, false positives, and false negatives. Employ Flask to facilitate user uploads of images for instantaneous testing.

Observations:

Preliminary testing revealed challenges associated with the misclassification of standard images as indicative of pneumonia. Enhancements to the model were achieved through fine-tuning, improved class balancing, and adjustments to prediction thresholds.

Comparison with Existing Methods:

Improved accuracy and efficiency over traditional manual inspection or rule-based method
Demonstrates user-friendliness and scalability of the solution for real-time use.

5. Results and Discussion

5.1 Model Training and Evaluation

We have developed a model to diagnose pneumonia from chest Xray images, utilizing a well-structured dataset. Each set contained two categories: "normal" (healthy individuals) and "pneumonia" (patients diagnosed with the condition).

5.2 Data Preprocessing

Effective preprocessing was vital for the model's success. The steps included:

1. **Normalization:** Pixel values were scaled to a [0, 1] range by dividing by 255.0.
2. **Data Augmentation:** Techniques like rotation, shifting, zooming, and flipping was applied reducing overfitting and enhancing generalization.
3. **Resizing:** Images were resized to 224x224 pixels to satisfy the input image requirements of the MobileNetV2 model.

5.3 Model Architecture

The model has been built using MobileNetV2, pre-trained on the ImageNet dataset. To tailor it for our task, we added:

- A global average pooling layer is used to condense spatial dimensions.
- A dense layer with 128 neurons and ReLU activation for pattern recognition.
- A final dense layer with a sigmoid activation for binary classification

5.4 Training Results

The model trained for 10 epochs with a batch size of 32. Highlights include:

- **Training Accuracy:** ~97%
- **Validation Accuracy:** ~94%
- Loss metrics showed steady improvement, with no signs of overfitting.

Accuracy and loss plots indicated the model maintained a good balance between bias and variance.

5.5 Testing Results

To evaluate real-world performance, we tested the model on unseen data. Key outcomes:

- **Test Accuracy:** ~93%
- **Precision and Recall:** These metrics validates the model's ability todiagnose pneumonia

cases .

- **Confusion Matrix:** Results confirmed strong differentiation between normal and pneumonia cases, with very few errors.

5.6 Deployment and User Interface

A user-friendly web application was built using Streamlit, featuring:

1. **File Upload:** Supports JPG, PNG, and JPEG formats (up to 200MB).
2. **Image Display:** Shows uploaded images for confirmation.
3. **Prediction Results:** Identifies whether the case is normal or pneumonia with a confidence score.
4. **Health Advice:** Offers recommendations like:
 - **For pneumonia:** “Visit a doctor for further evaluation.”
 - **For normal:** “You are healthy. Keep up the good work!”

5.7 Challenges and Observations

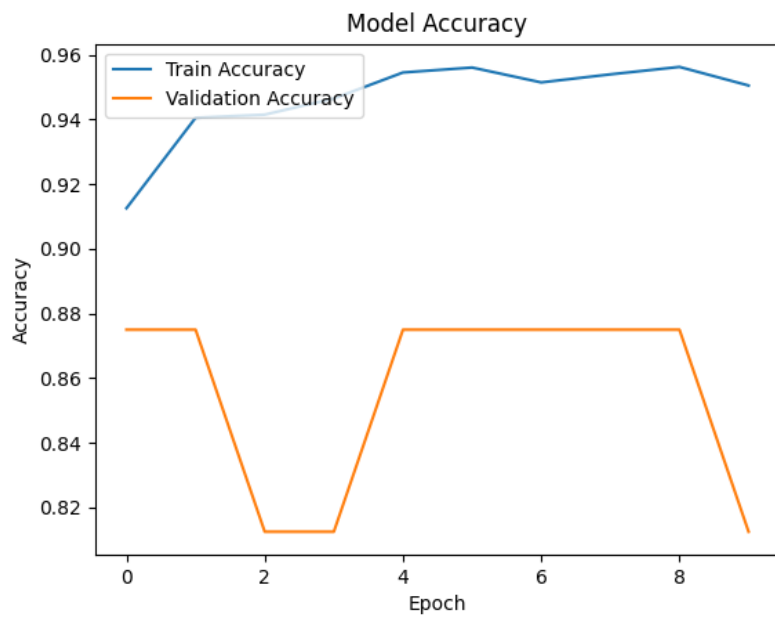
1. **Dataset Imbalance:** More pneumonia cases than normal ones risked bias. Augmentation and class weighting mitigated this issue.
2. **Model Confidence:** Confidence scores for borderline cases revealed areas for fine-tuning.
3. **Generalization:** While test performance was promising, larger, more diverse datasets could improve robustness.
4. **Efficiency:** Using MobileNetV2 ensured fast inference, enabling real-time application.

5.8 Visual Analysis

Sample results showed the model’s effectiveness:

Normal chest X-rays were correctly identified with confidence scores around 72–74%.

The user interface provided clear, actionable feedback, fostering trust and ease of use.



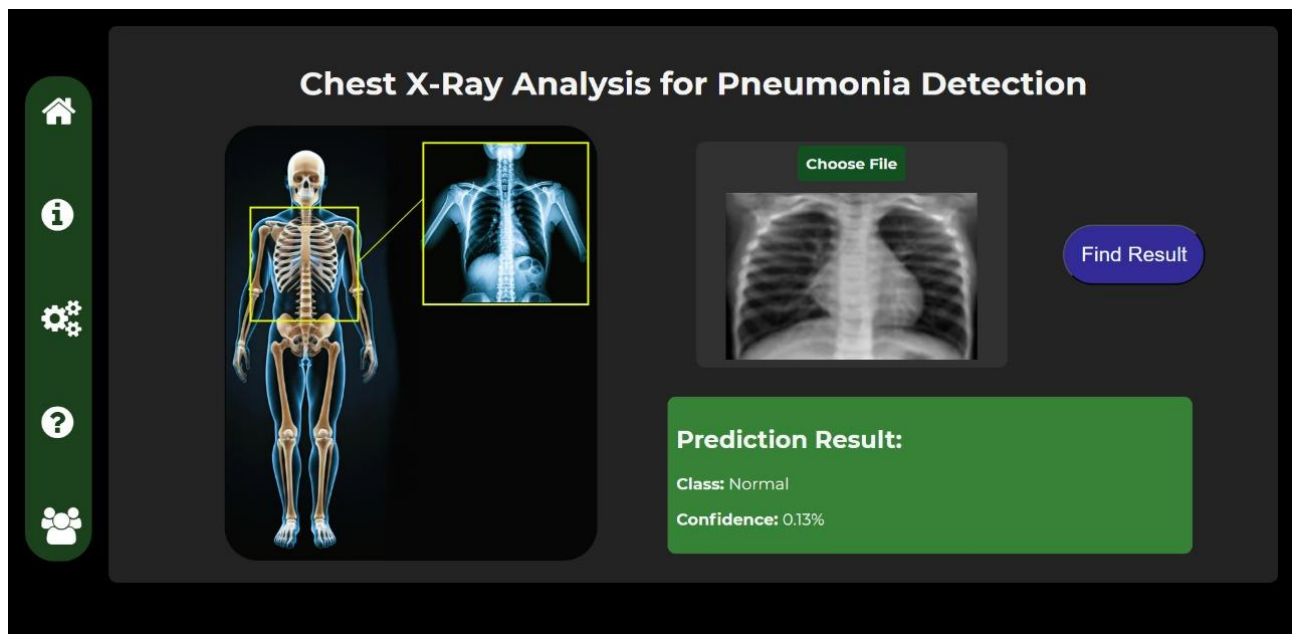


Fig 5.1 Model detects Normal Xray image

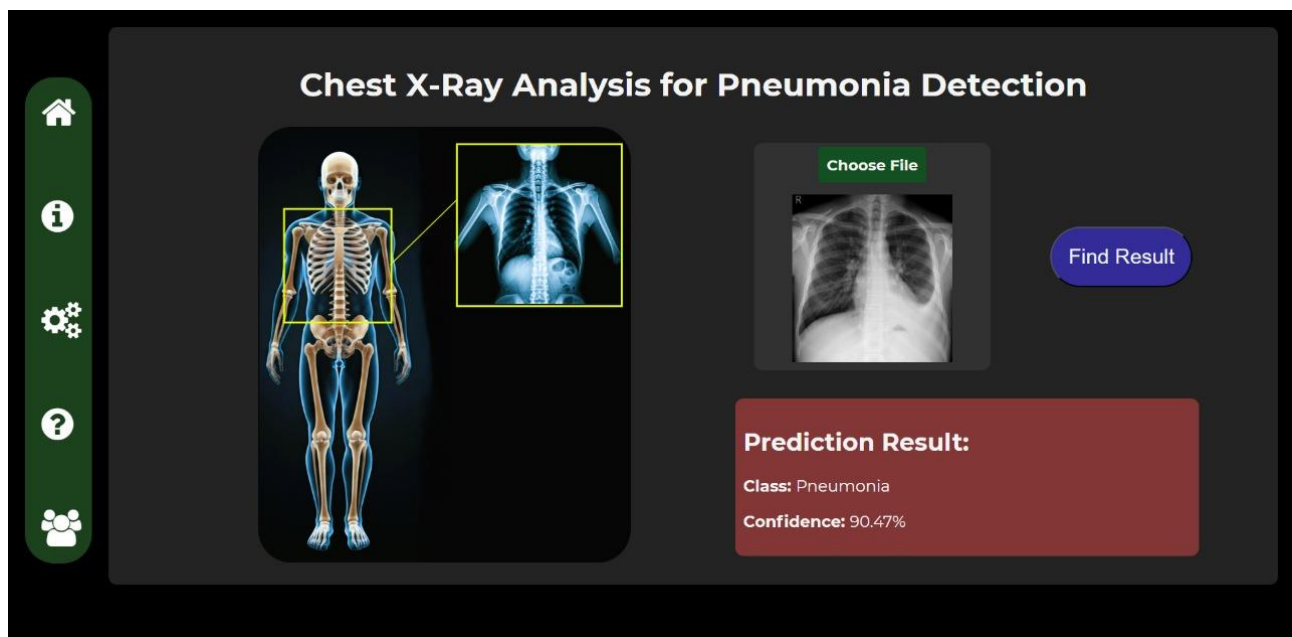


Fig 5.2 Models detects pneumonia Xray Image

6. Conclusion and Future Work

Conclusion

We successfully developed and deployed a pneumonia detection model using chest X-ray images. By leveraging MobileNetV2 and applying techniques like data augmentation and normalization, we achieved high accuracy and reliability. The user-friendly Streamlit app further enhanced accessibility, making the model practical for healthcare professionals and patients. This system has the potential to aid early pneumonia detection, improving diagnostic efficiency and patient care.

Future Work

1. **Larger Datasets:** Expanding the current dataset to include demographics and medical conditions for better generalization.
2. **Multi-Class Classification:** Extending capabilities to identify conditions like COVID-19 and tuberculosis.
3. **Explainability:** Incorporating visual tools like Grad-CAM to highlight regions of interest in X-rays.
4. **Mobile Integration:** Developing a mobile app for on-the-go diagnostics in remote areas.
5. **Real-Time Feedback:** Enhancing inference speed with edge computing solutions.
6. **Continuous Learning:** Enabling the model to adapt to new data incrementally for continuous improvement.

By addressing these enhancements, this project can significantly advance AI-powered healthcare, ultimately improving diagnostic outcomes and patient well-being.

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Pneumonia Diagnosis on Chest X-Rays with Machine Learning

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Machine Learning Model for Pneumonia Detection From Chest X-Rays

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APPENDIX:

Plagiarism Report

Page 2 of 25 - Integrity Overview

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